ECO3121 Tutorial 9

Fall 2022

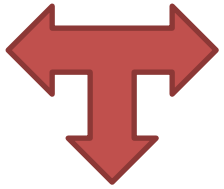
Yichen's Teaching Notes

Nov. 9

MERGE – EXAMPLE 1 (one to one)

mydata1

	country	year	y	y_bin	x1	x2	x3
1	A	2000	1343	1	.28	-1.11	.28
2	A	2001	-1900	0	.32	-.95	.49
3	A	2002	-11	0	.36	-.79	.7
4	A	2003	2646	1	.25	-.89	-.09
5	B	2000	-5935	0	-.08	1.43	.02
6	B	2001	-712	0	.11	1.65	.26
7	B	2002	-1933	0	.35	1.59	-.23
8	B	2003	3073	1	.73	1.69	.26
9	C	2000	-1292	0	1.31	-1.29	.2
10	C	2001	-3416	0	1.18	-1.34	.28
11	C	2002	-356	0	1.26	-1.26	.37
12	C	2003	1225	1	1.42	-1.31	-.38



mydata2

	country	year	x4	x5	x6	order
1	A	2000	10	1	9	1
2	A	2001	7	1	9	2
3	A	2002	7	9	4	3
4	A	2003	1	2	3	4
5	B	2000	0	5	6	5
6	B	2001	5	8	5	6
7	B	2002	9	4	5	7
8	B	2003	1	5	1	8
9	C	2000	4	5	4	9
10	C	2001	6	9	6	10
11	C	2002	6	5	3	11
12	C	2003	7	3	3	12

```
merge 1:1 country year using mydata2
```

```
Result                           # of obs.
-----------------------------------------
not matched                             0
matched                                12   (_merge==3)
-----------------------------------------
```

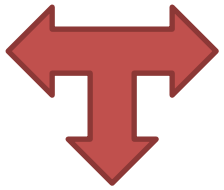
- It is equivalent to **adding columns**
- Make sure one dataset is loaded into Stata (in this case mydata1), then use merge.
- Make sure to map where the using data is located (in this case mydata2, for example "c:\folders\data\mydata2.dta")*.

	country	year	y	y_bin	x1	x2	x3	x4	x5	x6	order	_merge
1	A	2000	1343	1	.28	-1.11	.28	10	1	9	1	matched (3)
2	A	2001	-1900	0	.32	-.95	.49	7	1	9	2	matched (3)
3	A	2002	-11	0	.36	-.79	.7	7	9	4	3	matched (3)
4	A	2003	2646	1	.25	-.89	-.09	1	2	3	4	matched (3)
5	B	2000	-5935	0	-.08	1.43	.02	0	5	6	5	matched (3)
6	B	2001	-712	0	.11	1.65	.26	5	8	5	6	matched (3)
7	B	2002	-1933	0	.35	1.59	-.23	9	4	5	7	matched (3)
8	B	2003	3073	1	.73	1.69	.26	1	5	1	8	matched (3)
9	C	2000	-1292	0	1.31	-1.29	.2	4	5	4	9	matched (3)
10	C	2001	-3416	0	1.18	-1.34	.28	6	9	6	10	matched (3)
11	C	2002	-356	0	1.26	-1.26	.37	6	5	3	11	matched (3)
12	C	2003	1225	1	1.42	-1.31	-.38	7	3	3	12	matched (3)

MERGE – EXAMPLE 2 (one to one)

mydata1

	country	year	y	y_bin	x1	x2	x3
1	A	2000	1343	1	.28	-1.11	.28
2	A	2001	-1900	0	.32	-.95	.49
3	A	2002	-11	0	.36	-.79	.7
4	A	2003	2646	1	.25	-.89	-.09
5	B	2000	-5935	0	-.08	1.43	.02
6	B	2001	-712	0	.11	1.65	.26
7	B	2002	-1933	0	.35	1.59	-.23
8	B	2003	3073	1	.73	1.69	.26
9	C	2000	-1292	0	1.31	-1.29	.2
10	C	2001	-3416	0	1.18	-1.34	.28
11	C	2002	-356	0	1.26	-1.26	.37
12	C	2003	1225	1	1.42	-1.31	-.38



mydata3

	country	year	x4	x5	x6	order
1	A	2000	10	1	9	1
2	A	2001	7	1	9	2
3	A	2002	7	9	4	3
4	A	2003	1	2	3	4
5	B	2000	0	5	6	5
6	B	2001	5	8	5	6
7	B	2002	9	4	5	7
8	B	2003	1	5	1	8

`merge 1:1 country year using mydata3`

```
Result                           # of obs.
-----------------------------------------
not matched                             4
    from master                         4  (_merge==1)
    from using                          0  (_merge==2)

matched                                 8  (_merge==3)
-----------------------------------------
```

- Unmatched data is set to missing. If you want to keep only matched data, you can type: **`keep if _merge==3`**
- or: **`drop if _merge==2`**

	country	year	y	y_bin	x1	x2	x3	x4	x5	x6	order	_merge
1	A	2000	1343	1	.28	-1.11	.28	10	1	9	1	matched (3)
2	A	2001	-1900	0	.32	-.95	.49	7	1	9	2	matched (3)
3	A	2002	-11	0	.36	-.79	.7	7	9	4	3	matched (3)
4	A	2003	2646	1	.25	-.89	-.09	1	2	3	4	matched (3)
5	B	2000	-5935	0	-.08	1.43	.02	0	5	6	5	matched (3)
6	B	2001	-712	0	.11	1.65	.26	5	8	5	6	matched (3)
7	B	2002	-1933	0	.35	1.59	-.23	9	4	5	7	matched (3)
8	B	2003	3073	1	.73	1.69	.26	1	5	1	8	matched (3)
9	C	2000	-1292	0	1.31	-1.29	.2	.	.	.	.	master only (1)
10	C	2001	-3416	0	1.18	-1.34	.28	.	.	.	.	master only (1)
11	C	2002	-356	0	1.26	-1.26	.37	.	.	.	.	master only (1)
12	C	2003	1225	1	1.42	-1.31	-.38	.	.	.	.	master only (1)

MERGE – EXAMPLE 3 (many to one)

mydata1

	country	year	y	y_bin	x1	x2	x3
1	A	2000	1343	1	.28	-1.11	.28
2	A	2001	-1900	0	.32	-.95	.49
3	A	2002	-11	0	.36	-.79	.7
4	A	2003	2646	1	.25	-.89	-.09
5	B	2000	-5935	0	-.08	1.43	.02
6	B	2001	-712	0	.11	1.65	.26
7	B	2002	-1933	0	.35	1.59	-.23
8	B	2003	3073	1	.73	1.69	.26
9	C	2000	-1292	0	1.31	-1.29	.2
10	C	2001	-3416	0	1.18	-1.34	.28
11	C	2002	-356	0	1.26	-1.26	.37
12	C	2003	1225	1	1.42	-1.31	-.38

mydata4

	country	x7
1	A	100
2	B	200
3	C	300

merge m:1 country using mydata4

```
Result                           # of obs.
-----------------------------------------
not matched                             0
matched                                12   (_merge==3)
-----------------------------------------
```

- Unmatched data is set to missing. If you want to keep only matched data, you can type: **keep if _merge==3**
- or: **drop if _merge==2**

	country	year	y	y_bin	x1	x2	x3	x7	_merge
1	A	2000	1343	1	.28	-1.11	.28	100	matched (3)
2	A	2001	-1900	0	.32	-.95	.49	100	matched (3)
3	A	2002	-11	0	.36	-.79	.7	100	matched (3)
4	A	2003	2646	1	.25	-.89	-.09	100	matched (3)
5	B	2000	-5935	0	-.08	1.43	.02	200	matched (3)
6	B	2001	-712	0	.11	1.65	.26	200	matched (3)
7	B	2002	-1933	0	.35	1.59	-.23	200	matched (3)
8	B	2003	3073	1	.73	1.69	.26	200	matched (3)
9	C	2000	-1292	0	1.31	-1.29	.2	300	matched (3)
10	C	2001	-3416	0	1.18	-1.34	.28	300	matched (3)
11	C	2002	-356	0	1.26	-1.26	.37	300	matched (3)
12	C	2003	1225	1	1.42	-1.31	-.38	300	matched (3)

Probit & Logit Model

Motivation (Drawbacks of Linear Probability Model)

1. Predicted probability may >1 or <0, which makes no sense in practice
2. Estimated partial effects could only be constant
3. Must have heteroskedasticity (y is a Bernoulli distribution rather than normal; error term is a function of x)

$$Var(y|\mathrm{x}) = P(y=1|\mathrm{x})\,[1 - P(y=1|\mathrm{x})]$$

← Heteroskedasticity

- We can use this to find the conditional variance of the error term

$$\begin{aligned}
Var\,(u_i|X_{1i},\cdots,X_{ki}) &= Var\,(Y_i - (\beta_0 + \beta_1 X_{1i} + \cdots \beta_k X_{ki})|\, X_{1i},\cdots,X_{ki}) \\
&= Var\,(Y_i|\, X_{1i},\cdots,X_{ki}) \\
&= Pr\,(Y_i = 1|\, X_{1i},\cdots,X_{ki}) \times (1 - Pr\,(Y_i = 1|\, X_{1i},\cdots,X_{ki})) \\
&= (\beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki}) \times (1 - \beta_0 - \beta_1 X_{1i} - \cdots - \beta_k X_{ki}) \\
&\neq \sigma_u^2
\end{aligned}$$

Probit & Logit Model

Probit & Logit:

1. Nonlinear in parameters
   - **Estimation:** Maximum Likelihood Estimation (MLE)
2. Do monotonic transformation to assure the range of predicted probability is within [0, 1]
   - **Monotonic transformation:** a way of transforming one set of numbers into another set of numbers in a way that the order of the numbers is preserved.
   - Using CDF!

Probit: cdf of **standard normal distribution**

$$P(Y = 1) = \Phi(\beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k)$$

Logit: cdf of **logistic distribution**

$$P(Y = 1) = \frac{e^{\beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n}}{1 + e^{\beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n}}$$

IV Regression – 2SLS

When we face endogeneity problems… (violation of MLR.4):

1. **Omitted variable bias**
2. **Reverse causality**
3. **Measurement error of x** (not of y!)
   - Self-reported error (e.g. when reporting income, personality,…)
   - Coding error (manually input error)

Two Stage Least Square Regression (2SLS):

1 $$X_i = \pi_0 + \pi_1 Z_i + v_i$$

2 $$Y_i = \beta_0 + \beta_1 \hat{X}_i + u_i$$

Required Conditions:

1. **Relevance:** corr(z, x) ≠ 0
2. **Exogeneity:** corr(z, u) = 0

* Why measurement error in the dependent variable would NOT cause bias:

When you want to estimate a simple model like

$$Y_i = \alpha + \beta X_i + \epsilon_i$$

and instead of the true Y_i you only observe it with some error $\widetilde{Y}_i = Y_i + \nu_i$ which is such that it is uncorrelated with X and ϵ, if you regress

$$\widetilde{Y}_i = \alpha + \beta X_i + \epsilon_i$$

your estimated β is

$$\begin{aligned}
\widehat{\beta} &= \frac{Cov(\widetilde{Y}_i, X_i)}{Var(X_i)} \\
&= \frac{Cov(Y_i + \nu_i, X_i)}{Var(X_i)} \\
&= \frac{Cov(\alpha + \beta X_i + \epsilon_i + \nu_i, X_i)}{Var(X_i)} \\
&= \frac{Cov(\alpha, X_i)}{Var(X_i)} + \beta \frac{Cov(X_i, X_i)}{Var(X_i)} + \frac{Cov(\epsilon_i, X_i)}{Var(X_i)} + \frac{Cov(\nu_i, X_i)}{Var(X_i)} \\
&= \beta \frac{Var(X_i)}{Var(X_i)} \\
&= \beta
\end{aligned}$$

IV Regression – 2SLS

Note: if we manually do the 2SLS, the standard errors are biased!

```
. reg rent hsngval_hat pcturban //second stage
```

Source	SS	df	MS		
Model	44819.9705	2	22409.9852	Number of obs	= 50
Residual	16423.1495	47	349.428713	F(2, 47)	= 64.13
				Prob > F	= 0.0000
				R-squared	= 0.7318
				Adj R-squared	= 0.7204
Total	61243.12	49	1249.85959	Root MSE	= 18.693

rent	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
hsngval_hat	.0031938	.000393	8.13	0.000	.0024032	.0039845
pcturban	-.5064118	.3049712	-1.66	0.103	-1.119935	.1071111
_cons	113.8143	12.99962	8.76	0.000	87.66247	139.9662

Alternatively, using `ivregress 2sls rent pcturban (hsngval = faminc)`

Instrumental variables (2SLS) regression

Number of obs	=	50
Wald chi2(2)	=	51.44
Prob > chi2	=	0.0000
R-squared	=	0.2887
Root MSE	=	29.517

rent	Coef.	Std. Err.	z	P>|z|	[95% Conf.	Interval]
hsngval	.0031938	.0006206	5.15	0.000	.0019775	.0044101
pcturban	-.5064118	.4815558	-1.05	0.293	-1.450244	.4374202
_cons	113.8143	20.52667	5.54	0.000	73.5828	154.0459

Instrumented: hsngval
Instruments: pcturban faminc